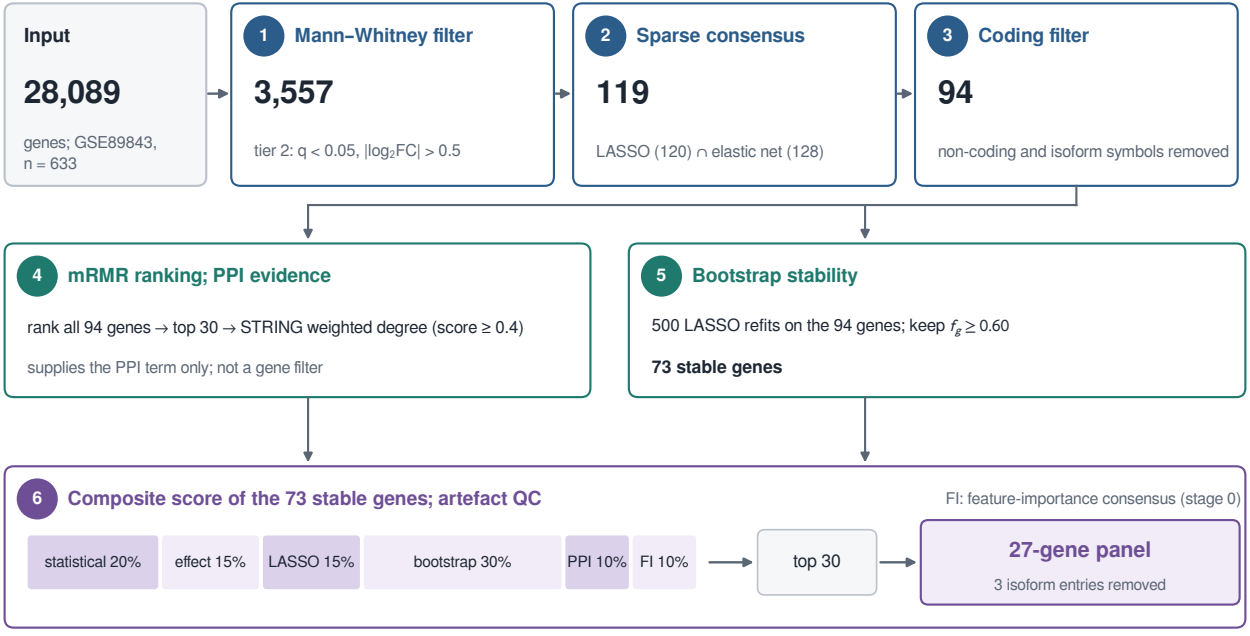
Input
28,089
genes; GSE89843, n = 633
1
Mann–Whitney filter
3,557
tier 2: q < 0.05, |log2FC| > 0.5
2
Sparse consensus
119
LASSO (120) ∩ elastic net (128)
3
Coding filter
94
non-coding and isoform symbols removed
4
mRMR ranking; PPI evidence
rank all 94 genes → top 30 → STRING weighted degree (score ≥ 0.4)
supplies the PPI term only; not a gene filter
5
Bootstrap stability
500 LASSO refits on the 94 genes; keep $f_g \geq 0.60$
73 stable genes
6
Composite score of the 73 stable genes; artefact QC
FI: feature-importance consensus (stage 0)
statistical 20%
effect 15%
LASSO 15%
bootstrap 30%
PPI 10%
FI 10%
top 30
27-gene panel
3 isoform entries removed